

Aditya Gautam

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Summary

Software Developer with **1.5+ years** building and operating **production backend services** for an EdTech platform serving **700K+ users** at **50K–70K peak concurrency**. Owns **systems design**, **REST API architecture**, and performance optimization across a 1M+ download application and multiple associated web platforms; delivered an **87.5% API latency reduction**. Strong in **object-oriented design**, distributed caching, queue-based processing, **automated testing**, and CI/CD. **400+** DSA problems solved in **Java**.

Education

Indian Institute of Information Technology (IIIT), Ranchi 2023 – 2027 (Expected)
Bachelor of Technology in Computer Science and Engineering CGPA: 7.6 / 10, No Backlogs
Award: 2nd Runner-Up, Reserve Bank of India National Quiz (2024).

Skills

Languages: Java, JavaScript, TypeScript, Python, SQL **Frontend:** React.js, Next.js (SSR/SSG/ISR), Redux Toolkit
Backend & Services: Node.js, Express.js, **REST API design**, WebSockets / Socket.io, OAuth 2.0, BullMQ (distributed job queues), Redis caching, rate limiting, background workers
Testing & Quality: **Playwright** (end-to-end), unit & integration testing (Jest), API testing (Postman), **automated test suites in CI** (GitHub Actions, per push/PR), peer code review
Cloud, Data & DevOps: AWS (S3, CDN), Docker, **CI/CD** (GitHub Actions), PostgreSQL, MongoDB, Redis, Firebase, Git-based workflows
Core CS: Object-Oriented Design, Data Structures & Algorithms, System Design (HLD/LLD), Operating Systems, DBMS, Computer Networks, Distributed Systems, Agile / Scrum

Experience

Software Developer, TeachMantra (Target Board) – Remote, India Jan 2026 – Present

- Own **systems design, backend structuring, and optimization** across the flagship application (**1M+ Play Store downloads, 700K+ students**) and multiple associated web platforms – shipping features end-to-end from design through testing, deployment, and **live production operation**, via sprint planning, design review, and **peer code review**.
- Reduced API response times from **800ms to 100ms (87.5% improvement)** at **50K–70K peak concurrent users** via query optimization, **Redis-backed caching layers**, request throttling, and debouncing – using latency and error metrics to locate bottlenecks and verify each change in production.
- Architected the **20+ module admin service platform** serving as the operational control plane for the product – **RBAC**, test management, stream configuration, payments, and push notifications – backed by **REST APIs**, role-scoped authorization, and **unit-tested** service logic.
- Built and operate the platform's **real-time service layer** – live-class delivery via third-party streaming integration, **Socket.io/WebSocket** chat sustaining tens of thousands of concurrent sessions with reconnection and session-state handling, and **Firebase push** fan-out to 700K+ users.
- Built an automated **payment reconciliation pipeline** (UTR matching with OCR verification) with the finance team, replacing a manual workflow; monitor service health and triage **live-site issues** in production.

Full Stack Developer Intern, ExamAdda – Remote, India June 2025 – Jan 2026

- **Proposed and led** the refactor of a monolithic application into **5 independently deployable modules** (admin panel, technical docs, current affairs, DSA visualizer, main app), improving maintainability and release velocity.
- Cut average page load from **3.5s to under 200ms** and platform quality scores from **40–50 to 90+** via SSG/ISR rendering, caching, and **CDN** delivery; built an in-browser **code execution service** with queue-based request management.

Projects

Dispatch – Queue-Driven Outreach Service *Node.js, BullMQ, Redis, PostgreSQL, Docker* GitHub

- Engineered a **distributed, queue-based backend service** for scheduled high-volume email delivery, using **BullMQ over Redis** for asynchronous job processing with retries, **exponential backoff**, and idempotent job keys preventing duplicate sends.
- Implemented **OAuth 2.0** with encrypted refresh-token storage and per-account **rate limiting** to stay within provider quotas, using least-privilege scopes and secure credential handling.
- Modelled each recipient as a **state machine persisted in PostgreSQL** with reply-detection logic that halts an in-flight sequence on response; containerized with **Docker**.

Fortune First – Portfolio, Finance Management *Playwright, Jest, GitHub Actions, Node.js* GitHub

- Designed a reusable **automated test framework – end-to-end flows** (Playwright), API/integration, and unit tests with shared fixtures and page objects – wired into **GitHub Actions on every push/PR**, gating merges on a green run.